

CP221: IP 1

TEST ONE 2024

Question One

Write a program that displays, validates, and processes a form intended for user registration. User registration captures the following details: username, password, gender (Male or Female), marital status (Single or Married), and preferences (Music, football, netball, volleyball). The form should be validated to prevent submission of empty fields.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Registration</title>
</head>
<body>
  <form id="registrationForm">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required><br><br>

    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required><br><br>

    <label for="gender">Gender:</label>
    <input type="radio" id="male" name="gender" value="Male" required> Male
    <input type="radio" id="female" name="gender" value="Female" required>
Female<br><br>

    <label for="maritalStatus">Marital Status:</label>
    <input type="radio" id="single" name="maritalStatus" value="Single" required>
Single
    <input type="radio" id="married" name="maritalStatus" value="Married"
required> Married<br><br>

    <label for="preferences">Preferences:</label><br>
```

```
        <input type="checkbox" id="music" name="preferences" value="Music"> Music<br>
        <input type="checkbox" id="football" name="preferences" value="Football">
Football<br>
        <input type="checkbox" id="netball" name="preferences" value="Netball">
Netball<br>
        <input type="checkbox" id="volleyball" name="preferences" value="Volleyball">
Volleyball<br><br>

        <button type="submit">Register</button>
    </form>

    <script>
        document.getElementById('registrationForm').addEventListener('submit',
function(event) {
            const username = document.getElementById('username').value;
            const password = document.getElementById('password').value;
            const gender = document.querySelector('input[name="gender"]:checked');
            const maritalStatus =
document.querySelector('input[name="maritalStatus"]:checked');
            const preferences =
document.querySelectorAll('input[name="preferences"]:checked');

            if (!username || !password || !gender || !maritalStatus ||
preferences.length === 0) {
                event.preventDefault();
                alert('Please fill all the fields.');
```

Question Two

Write the CSS rules that do the following:

```
a) Round the corners of the button with class="button" to 2px
.button {
    border-radius: 2px;
}

b) Add black shadows to the button with type="button" to 2px
button[type="button"] {
    box-shadow: 2px 2px 2px black;
}

c) Remove an element with id="element"
#element {
    display: none;
}
```

Question Three

a) Write a JavaScript function, which receives an array of numbers as an argument and returns the product of the numbers in the array.

```
function productOfArray(numbers) {
    let product = 1;
    for (let i = 0; i < numbers.length; i++) {
        product *= numbers[i];
    }
    return product;
}
```

b) Use the JavaScript Document Object Model (DOM) to create a paragraph and add it to the HTML document body. The paragraph displays the message "Welcome to programming".

```
const paragraph = document.createElement('p');
paragraph.innerHTML = 'Welcome to programming';
document.body.appendChild(paragraph);
```

c) Write a JavaScript function that deletes an HTML element with id="item" from the document when an HTML button with id="button" is clicked.

```
document.getElementById('button').onclick = function() {
    const item = document.getElementById('item');
    item && item.parentNode.removeChild(item);
};
```

d) Write a JavaScript function that receives two numbers from a user through Browser Object Model (BOM) popups and displays the summation of the numbers through BOM popups.

```
function sumTwoNumbers() {
    const num1 = parseFloat(prompt('Enter the first number:'));
    const num2 = parseFloat(prompt('Enter the second number:'));
    alert('The sum is: ' + (num1 + num2));
}
```